



INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

ASSIGNMENT

Assignment Title: Networking Fundamentals

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SECTION A

- 1) Flat Addressing: The flat addressing mainly uses the intranet (local area network), it is situated on the OSI layer 2 uses a single-level structure where each device has a unique identifier, it is statistically assigned based on the next available number. An example is the mac address.

Hierarchical Addressing: This uses the internet itself, it is located at layer 3, like IP addresses, organizes devices into networks and sub-networks, making routing more efficient, it is assigned dynamically based on location.

- 2) IANA (Internet Assigned Numbers Authority) oversees global IP address allocation and delegates blocks to Regional Internet Registries (RIRs). RIRs then distribute IP addresses to ISPs and organizations within their regions.
- 3) For an internet to be routable, we require:
- i) An IP Address
 - ii) A Subnet Mask
 - iii) A Default Gateway
 - iv) A DNS Address (only for websites lookup)

- 4) A static IP is manually assigned by the administrator and doesn't change; it is ideal for servers.

A dynamic IP is automatically assigned by Dynamic Host Configuration Protocol (DHCP) which requires less human intervention and change each time the device is connected to the internet, it is suitable for everyday devices like laptops.

- 5) Classful IP Addressing: The classful IP addressing is further divided into five (5) classes, the multicast (A, B, C, D) and the reserve (E) and does not need to send information to the subnet, it also makes sure that all networks are the same size and also have the same subnet mask.

Classless IP Addressing: The Classless addressing, also known as classless inter-domain routing (cidr) is more efficient because it reduces wasted IPs. This IP addressing sends information to the subnet which makes it vulnerable to attack.

SECTION B

IP Address	Class	Network Address	Broadcast Address	Default Subnet Mask
10.50.100.10	A	10.0.0.0	10.255.255.255	255.0.0.0
172.16.0.25	B	172.16.0.0	172.16.255.255	255.255.0.0
192.168.1.150	C	192.168.1.0	192.168.1.255	255.255.255.0
91.118.125.2	A	91.0.0.0	91.255.255.255	255.0.0.0
203.0.113.45	C	203.0.113.0	203.0.113.255	255.255.255.0

SECTION C

16. Special Purpose of 127.0.0.1 Address is the loopback/lookup address or testing. It lets a device send data to itself, often used for testing.
17. 169.254.10.1/16 means the device couldn't get an IP from a DHCP server. It self-assigned an address to communicate locally.
18. The default gateway must be on the same network with the host so the host can reach it directly. Otherwise, packets won't be delivered.
19. For 192.168.50.0/24:
Total addresses: $2^8 = 256$
Usable: $256 - 2 = 254$
20. For 198.51.100.0/27:
Subnet mask: **255.255.255.224**
Usable hosts: $2^5 - 2 = 30$
21. Using 192.168.10.0/24 for 12 devices is inefficient. It supports 254 hosts, so most addresses go unused. A smaller subnet like /28 would be better.
22. False. 224.0.0.10 is a multicast address, reserved for group communication. It can't be assigned to a single device like a laptop.
23. Yes, it will use the gateway. The destination (10.10.11.10) is outside the sender's subnet (10.10.10.0/24), so routing is needed.

SECTION D: BONUS SECTION

The IP (172.20.150.5) and gateway (172.21.150.1) are on different networks. With a subnet mask of 255.255.0.0, they should be in the same 172.20.0.0 network.